Resource Allocation and Load Balancing for Beam Hopping Scheduling in Satellite-Terrestrial Communications: A Cooperative Satellite Approach

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***Abstract*— Satellite-terrestrial communications based on mega low-Earth orbit (LEO) constellations enable extensive coverage and high data rates. However, the communication performance is significantly impacted by the non-uniform traffic distribution and the substantial interference caused by dense satellites. Therefore, in this paper, a multi-satellite cooperation architecture for satellite-terrestrial communications is proposed in LEO satellite constellations. Specifically, beam hopping (BH) and resource allocation enable a flexible solution for the non-uniform geographical distribution of communications, and are optimized to improve communication performance and avoid both intra- and inter-satellite interference in satellite-terrestrial communications. Moreover, load balancing via inter-satellite link (ISL) is implemented to further enhance the communication performance. Consequently, the overall problem for maximizing the network throughput and ensuring the latency metric is formulated, and then decomposed into three sub-problems: deep reinforcement learning (DRL) based BH scheduling, resource allocation with multi-satellite cooperation, and load balancing via ISLs. Specifically, DRL is utilized to determine the real-time BH pattern, and the resource allocation among beams is implemented by the majorization-minimization algorithm. Furthermore, for varying input traffic loads, different objectives for load balancing are established and solved by quadratic transformation and hybrid block successive approximation algorithm. Simulation results demonstrate that the proposed method outperforms other existing methods. Meanwhile, it obtains an 18.45% improvement in throughput compared with the benchmark without optimization, while the latency metric for satellite-terrestrial communications is also reduced by the proposed method.**

***Index Terms*— Satellite-terrestrial communications, multi-satellite cooperation, beam hopping, resource allocation, load balancing, inter-satellite link.**

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I. Introduction

THE sixth-generation wireless networks aim to achieve wide area coverage, however, the economic cost and construction challenges pose difficulties in covering extensive underdeveloped areas through ground base stations [2]. Therefore, satellite-terrestrial communications, covering most of the Earth's surface, serve a significant role in the coverage of remote areas for next-generation wireless communications [3]. However, the high orbit altitude results in increased latency of satellite-terrestrial communications, while the considerable construction cost of traditional satellite constellations also diminishes its application scope [4]. Recently, the low Earth orbit (LEO) satellite constellations with low construction cost, exemplified by Starlink and OneWeb, have attracted attention due to the low latency and high throughput [5].

However, the geographical distribution of traffic demands is notably non-uniform, and the high-speed motion of LEO satellites results in the time-varying characteristic of satellite-terrestrial communications. To address these challenges, resource allocation among beams in multi-beam satellites is crucial [6]. Consequently, the genetic algorithm was employed for power allocation among beams in [7], while a simulated annealing method was utilized to optimize both power and spectrum resources in [8]. Moreover, the deep reinforcement learning (DRL) was also introduced in [9] to improve the coverage performance of multi-beam satellites.

Nevertheless, these approaches retain a fixed pointing direction of the beams, and the freedoms in time and space are not fully exploited. Consequently, beam hopping (BH), which enables dynamic adjustment of satellite beam pointing, emerges as a flexible solution [10], and is adopted by the DVB-S2X standard [11]. Moreover, the network performance with BH was explored in [12] through queuing theory, while enhancements in energy efficiency for satellite-terrestrial

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communications were achieved by selecting proper BH pattern [13]. Besides, hybrid precoding based on BH was studied in [14] and [15] to mitigate the intra-satellite interference. Moreover, a BH scheduling method was proposed to cover the cells by maximizing the signal-to-interference plus noise ratio (SINR) [16]. However, these methods for the BH scheduling neglect to match the traffic demands. Combined with the non-uniform input traffic, game theory was utilized to maximize the throughput with BH in [17], while matching theory [18] was also employed. Additionally, the BH pattern selection was formulated as a mixed integer programming problem, and non-orthogonal multiple access among beams was incorporated to further enhance the throughput [19]. However, these methods face challenges in time-varying scenarios, because they focus solely on meeting the total traffic demand and overlook the randomness of traffic arrivals, while also having high computational complexity. In contrast, genetic algorithms were utilized in [20] to determine BH patterns in real-time, and the user service weight gain (USWG) was evaluated in [21] to determine the priority of each cell. Moreover, considering the long-term network performance, DRL can effectively handle dynamic scenarios and was adopted for optimizing BH patterns in [22]. Besides, to tackle the exponential growth in feasible BH patterns with the increase in beams and coverage cells, a multi-agent DRL method was further proposed [23].

Although previous research has made some progress for the BH scheduling in satellite-terrestrial communications, these studies are based on the optimization of BH for a single satellite. However, the complexity of the problem notably increases in multi-satellite scenarios, and the inter-satellite interference significantly reduces the throughput of satellite-terrestrial communications. Consequently, these studies cannot be directly utilized for multi-satellite scenarios. Moreover, the dense LEO satellite constellations enable the feasibility of multi-satellite cooperation. Therefore, the BH with multi-satellite cooperation holds substantial importance. For the multi-satellite scenarios, the research in [24] aims to address the non-uniform traffic distribution, and a fixed pre-scheduling BH time plan was implemented. However, this method schedules BH patterns for a long period of time based on the historical traffic statistics, while the rapid movement of LEO satellites results in highly time-varying characteristics, rendering the fixed BH time plan inadequate for time-varying conditions [25]. Moreover, a multi-agent DRL method was implemented to select the BH patterns among multiple satellites in [26]. Nevertheless, all beams occupy the same frequency band, resulting in severe interference among beams. To mitigate this, an adjacent beam avoidance requirement was imposed in [26]. However, this approach sacrifices the flexibility of BH and leads to performance loss when covering large areas with high load [15]. Meanwhile, although this method is feasible with the parameters given in [26], as the number of coverage cells and equipped beams per satellite increases, the action space will become extremely large. Additionally, it does not consider multi-satellite cooperation to enhance the overall performance of the satellite network.

Besides, the performance of satellite communications can be effectively enhanced through inter-satellite links (ISLs) [27], [28]. The collaborative data download through the ISLs in radio frequency among LEO satellites was studied in [29]. Recently, the application of laser ISLs further improved the data transmission rates among satellites [30]. Consequently, real-time load balancing among satellites can be potentially achieved via ISLs to adapt to the non-uniform traffic distribution. However, the aforementioned literature [24], [26] determined cell-satellite associations through the control center to realize load balancing, resulting in significant update overhead.

Therefore, existing schemes still have shortcomings in BH scheduling generality, resource allocation with multi-satellite cooperation, and load balancing via ISLs, which impair the performance of satellite-terrestrial communications. To overcome these challenges, we make several innovative contributions in this paper:

1) A general and effective multi-satellite cooperation architecture with BH is proposed for satellite-terrestrial communications, and the throughput and latency metric of the satellite network is optimized. The overall optimization problem is formulated and then divided into three sub-problems, including scheduling of the BH, resource allocation among beams, and load balancing via ISLs. Specifically, the multi-agent DRL is applied for each satellite to adapt to the non-uniform and time-varying traffic distribution. Simulation results show that compared with the benchmark, the proposed method improves the throughput of satellite-terrestrial communications by 18.45%, while the latency metric is also ensured. Additionally, compared to the existing methods, the proposed one can be accommodated with a larger number of coverage cells and equipped beams per satellite.
2) The resource allocation among beams is implemented based on real-time multi-satellite cooperation, which effectively eliminates both the intra- and inter-satellite interference in multi-satellite BH scenarios. By relaxing the constraints and utilizing the majorization-minimization (MM) algorithm, the non-convex resource allocation problem is transformed into the iteration of a series of linear programming problems, which converge rapidly. Additionally, simulation results show that resource allocation increases the flexibility of the BH scheduling.
3) To the best of our knowledge, the load balancing among multiple satellites is first implemented via ISLs in this paper to enhance the network performance. Specifically, load balancing aims to decrease the latency metric of the network when the input traffic is low. Otherwise, the goal shifts to equalizing the load distribution among satellites when the input traffic surpasses the network capacity. Therefore, the objectives of load balancing are determined under different situations, and the corresponding optimization problems are formulated.

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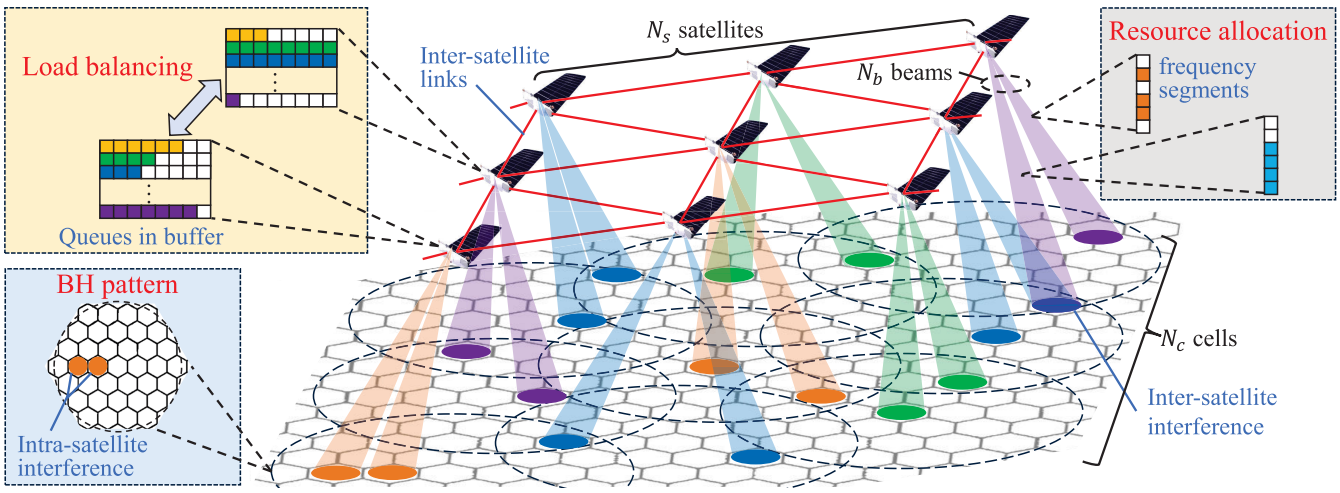

Fig. 1. The multi-satellite cooperation architecture with BH for satellite-terrestrial communications with load balancing.

The paper is structured as follows. Section II presents the framework of multi-satellite BH and the traffic distribution. Moreover, the optimization problem with various constraints is formulated in Section III, and the overall optimization problem is decomposed into three sub-problems on different time scales. In Section IV, the multi-agent DRL is adopted to determine the BH pattern to maximize the throughput and reduce the latency metric. Section V introduces the MM algorithm to optimize the non-convex problem for resource allocation. Meanwhile, in Section VI, from a long-term perspective, different objectives of load balancing are established and solved under various traffic loads. The simulation results are presented and analyzed in Section VII. Finally, the paper concludes with a summary in Section VIII.

II. System Model

The satellite-terrestrial communications based on the multi-satellite framework with BH and the frequency division channels for BH are introduced in this section, as well as the traffic distribution and load balancing among cells.

A. Framework of Multi-Satellite Beam Hopping

The downlink from multiple LEO satellites to terrestrial users is investigated in this paper, as shown in Fig. 1. A constellation comprising N_s LEO satellites is employed to provide coverage for N_c ground cells. It is assumed that the set of cells encompassed within the coverage area of satellite s is expressed as $\Omega(s)$, and each satellite can cover an identical quantity of cells $|\Omega(s)|$. Concurrently, a single cell may fall under the coverage of multiple satellites simultaneously, and thus the set of satellites with the potential to cover the i-th cell is signified as $\Phi(i)$. Moreover, each satellite is equipped with N_b beams. Due to the hardware constraints on the satellites and the demand to mitigate the interference among beams, $N_b < |\Omega(s)|$, and the beams are scheduled to hop between distinct time slots. Therefore, a well-designed BH pattern can satisfy the non-uniform service demands of each cell. Particularly, in time slot n, the BH pattern for satellite s can be represented as

$$\boldsymbol{a}_s(n) = \{a_{s,1}(n), a_{s,2}(n) \cdots, a_{s,N_b}(n)\}, \tag{1}$$

where $a_{s,k}(n)$ signifies the cell served by the k-th beam for satellite s, and manifestly $a_{s,k}(n) \in \Omega(s)$.

For a multi-beam satellite network, the achievable data rate of a beam is influenced not only by thermal noise but also by the interference from other beams. The channel gain from the j-th beam of satellite r to the cell illuminated by the k-th beam of satellite s is defined as $h_{s,k}^{r,j}$, and the vector $\boldsymbol{h}$ is composed of all $h_{s,k}^{r,j}$. To ensure the quality of service within each cell, the attenuation of the received signal strength at the cell edge relative to the center position should be within 3 dB [24]. Therefore, the beam interference to nearby cells remains a considerable concern and necessitates to be addressed in the system design and optimization.

Moreover, multiple beams from different satellites probably serve the same cell simultaneously. For this issue, frequency division multiple access offers a flexible solution to mitigate the interference and accommodate the non-uniform distribution of traffic. Therefore, the entire available frequency spectrum is uniformly partitioned into N_L segments, each with a bandwidth of B_0. The variable $f_{s,k,l} \in \{0,1\}$ is defined to indicate the allocation of the l-th frequency segment to the k-th beam for satellite s, and $f_{s,k,l} = 1$ signifies that the corresponding frequency segment is utilized, otherwise the opposite, while the vector formed by all $f_{s,k,l}$ is denoted by $\boldsymbol{f}$.

The power allocated to each utilized frequency segment is identical across all beams, represented by P_b. Nevertheless, the power of a single satellite is constrained not to exceed P^{sat}, and concurrently, the total power across the satellite network is restricted to a maximum power of P^{tot}.

B. Traffic Distribution and Load Balancing

During each time slot whose duration is T_0, the arriving data packet is retained within the buffer on the satellite for transmission, and each data packet is assumed to be of fixed size M_0. Owing to the constraints of storage volume, the duration that each data packet is permitted to be reserved in

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the buffer cannot surpass its specified time-to-live threshold T_{th}, while exceeding ones will be discarded. Consequently, the queue for the i-th cell on satellite s at the start of time slot n can be characterized as

$$\boldsymbol{d}_{s,i}(n) = \left\{\psi_{s,i}^{1}(n), \psi_{s,i}^{2}(n), \cdots, \psi_{s,i}^{T_{th}}(n)\right\}, \tag{2}$$

where $\psi_{s,i}^{t}(n)$ denotes the amount of data packets that have been waited for $t-1$ time slots in the queue $\boldsymbol{d}_{s,i}(n)$, and $\psi_{s,i}^{1}(n)$ represents the count of the data packets arrived during time slot n. Therefore, the total volume of data that is queued for transmission to the i-th cell in the buffer on satellite s is given by

$$M_{s,i}(n) = M_0 \sum_{t=1}^{T_{th}} \psi_{s,i}^{t}(n). \tag{3}$$

Moreover, as defined in literature [22], [23], the queuing delay $\tau_{s,i}(n)$ for queue $\boldsymbol{d}_{s,i}(n)$ in time slot n can be calculated by the average number of waiting time slots as

$$\tau_{s,i}(n) = \frac{\sum_{t=2}^{T_{th}} (t-1)\psi_{s,i}^{t}(n)}{\sum_{t=2}^{t=T_{th}} \psi_{s,i}^{t}(n)}. \tag{4}$$

Meanwhile, $\mathbf{D}_s(n)$ is utilized to describe the status of all the queues in the buffer of satellite s, which can be expressed as

$$\mathbf{D}_s(n) = \left\{\boldsymbol{d}_{s,i}(n) | i \in \Omega(s)\right\}. \tag{5}$$

For the i-th cell, the packet arrivals per time slot adhere to a Poisson distribution with a rate denoted by Λ_i. In addition, given the short duration of a time slot, it is assumed that the statistical characteristics of traffic arrivals will not change drastically. Due to the random characteristics of network routing, packets intended for the i-th cell are distributed with equal probability to the satellites encompassed in $\Phi(i)$ [26], thereby dividing the input traffic data flow into $\Phi(i)$ equivalent Poisson streams, each with an identical rate $\lambda_i^- = \Lambda_i/|\Phi(i)|$ directed towards its respective satellite. However, the traffic demand exhibits a geographical imbalance, which can lead to significant disparities among satellites. Therefore, ISLs could be employed to facilitate load balancing among satellites. Upon achieving load balancing, the packet arrival rate for the i-th cell in satellite s is denoted as $\lambda_{s,i}$, and all $\lambda_{s,i}$ form the vector $\boldsymbol{\lambda}$. Moreover, the transmission rate of data packets $\Delta\lambda_i$ that will be transmitted via ISLs in order to realize load balancing for the i-th cell, which can be calculated as

$$\Delta\lambda_i = \frac{1}{2} \sum_{s \in \Phi(i)} |\lambda_{s,i} - \lambda_i^-|, \tag{6}$$

where $\sum_{s\in\Phi(i)} |\lambda_{s,i} - \lambda_i^-|$ represents the sum of the absolute changes in the arrival rate of each queue for the i-th cell after load balancing. Since this value is computed once at both the transmitter and receiver, the total data rate transmitted through the ISLs for load balancing of the i-th cell should be halved. Owing to the high data rate offered by laser links [30], it is feasible to achieve all inter-satellite data transmissions within a single time slot. Consequently, data packets are redistributed among satellites for load balancing within one additional time slot.

III. Problem Formulation

In this section, the performance metrics of the satellite network, including the throughput and latency metric, are defined and calculated, and the overall optimization problem is formulated with the objectives of maximizing the throughput and reducing the latency metric. Moreover, the overall problem is decomposed into three sub-problems on different time scales, including DRL-based BH scheduling, resource allocation among beams, and load balancing via ISLs.

A. Throughput and Latency Metric

Each beam experiences interference from other beams that share the same frequency segment. Therefore, given the BH pattern $\boldsymbol{a}_s(n)$ for each satellite, the interference encountered by the l-th frequency segment of the k-th beam on satellite s can be calculated as

$$I_{s,k,l} = \sum_{\substack{r=1,\\ r\neq s}}^{N_s} \sum_{j=1}^{N_b} f_{r,j,l} \left|h_{s,k}^{r,j}\right|^2 P_b + \sum_{\substack{j=1,\\ j\neq k}}^{N_b} f_{s,j,l} \left|h_{s,k}^{s,j}\right|^2 P_b, \tag{7}$$

which includes the inter-satellite interference formulated by the first term, and the intra-satellite interference represented by the latter one. Hence, considering the thermal noise, the SINR $\gamma_{s,k,l}(\boldsymbol{f}, P_b)$ in the l-th frequency segment of the k-th beam on satellite s can be expressed as

$$\gamma_{s,k,l}(\boldsymbol{f}, P_b) = \frac{f_{s,k,l} \left|h_{s,k}^{s,k}\right|^2 P_b}{n_0 B_0 + I_{s,k,l}}, \tag{8}$$

where n_0 is the spectral power density of the thermal noise. Then, the achievable data rate $\tilde{R}_{s,k}(n)$ of the k-th beam for satellite s is contingent upon the frequency segments allocated to the beam and the SINR. Therefore, the achievable data rate $R_{s,k,l}$ in the l-th frequency segment can be first calculated, and then be aggregated to obtain the whole achievable data rate $\tilde{R}_{s,k}(n)$, which is given by

$$\tilde{R}_{s,k}(n) = \sum_{l=1}^{N_L} R_{s,k,l} = \sum_{l=1}^{N_L} f_{s,k,l} B_0 \log_2(1 + \gamma_{s,k,l}(\boldsymbol{f}, P_b)). \tag{9}$$

Upon the BH pattern being determined, if the achievable data rate $\tilde{R}_{s,k}(n)$ of the k-th beam for satellite s is insufficient to clear the buffer corresponding to cell $a_{s,k}(n)$ within one time slot, the packets adhere to a first come first served (FCFS) protocol. Therefore, packets with longer wait times are prioritized for transmission, and the throughput can be expressed as

$$y_{s,k}(n) = \min\left\{\tilde{R}_{s,k}(n) T_0,\ M_{s,a_{s,k}(n)}(n)\right\}. \tag{10}$$

Moreover, the latency performance of the network is also a crucial metric, and $\Delta\tau_s(n)$ is introduced to represent the latency metric for the satellite s, which is given by

$$\Delta\tau_s(n) = (\overline{\tau_s}(n) - \underline{\tau_s}(n)) + \kappa\overline{\tau_s}(n), \tag{11}$$

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where $\overline{\tau_s}(n) = \max_{i\in\Omega(s)}\tau_{s,i}(n)$ and $\underline{\tau_s}(n) = \min_{i\in\Omega(s)}\tau_{s,i}(n)$ denote the maximum and minimum queuing delays of the queues on satellite s, respectively. The former term denotes the delay fairness among cells covered by satellite s [22], [23], which is the primary optimization objective for latency performance in this paper. However, focusing solely on fairness may lead to increased queuing delays in all cells. Therefore, the second term, involving the maximum delay $\overline{\tau_s}(n)$, is introduced. Moreover, to prioritize the delay fairness within the latency metric, the coefficient κ in the latter term is assigned a small value. Consequently, the latency metric $\Delta\tau_s(n)$ aims to ensure delay fairness while reducing the maximum delay.

B. Overall Objectives and Constraints

In satellite-terrestrial communications, the throughput and latency are basic evaluation metrics. Consequently, this study aims to enhance the long-term performance of the satellite network to maximize the throughput $\sum_{k=1}^{N_b} y_{s,k}(n)$, and minimize the latency metric $\Delta\tau_s(n)$ for each satellite to ensure the delay fairness and reduce the maximum delay. Therefore, the overall objective function for satellite s in time slot n can be expressed as

$$r_s(n) = \beta\frac{\sum_{k=1}^{N_b} y_{s,k}(n)}{\Upsilon_0} - (1-\beta)\frac{\Delta\tau_s(n)}{\Gamma_0}, \tag{12}$$

where the two terms denote the throughput and the latency metric of satellite s, respectively. Since the two terms have different units, the normalization constants Υ_0 and Γ_0 are introduced for a reasonable addition. Additionally, given that the two objectives may not always align, the trade-off coefficient $0 \le \beta \le 1$ is adopted [22], [23], [26]. Moreover, optimizations are performed at each time slot, forming a sequential decision process. Hence, the optimization problem can be formulated as

$$\text{(P0)}: \max_{\boldsymbol{a}_s(n), P_b, \boldsymbol{f}, \boldsymbol{\lambda}} \lim_{N\to+\infty} \frac{1}{N}\sum_{n=1}^{N}\sum_{s=1}^{N_s} r_s(n), \tag{13}$$

$$\text{s.t. } a_{s,k}(n) \in \Omega(s), \quad \forall\, s\in\mathcal{S},\ \forall\, k\in\mathcal{K}, \tag{13a}$$

$$a_{s,j}(n) \neq a_{s,k}(n), \quad \forall\, s\in\mathcal{S}, \forall\, j,k\in\mathcal{K},\ j\neq k, \tag{13b}$$

$$f_{s,k,l} \in \{0,1\}, \quad \forall\, s\in\mathcal{S},\ \forall\, k\in\mathcal{K}, \forall\, l\in\mathcal{L}, \tag{13c}$$

$$\sum_{s=1}^{N_s}\sum_{k=1}^{N_b}\sum_{l=1}^{N_L} f_{s,k,l}P_b \le P^{tot}, \tag{13d}$$

$$\sum_{k=1}^{N_b}\sum_{l=1}^{N_L} f_{s,k,l}P_b \le P^{sat}, \quad \forall\, s\in\mathcal{S}, \tag{13e}$$

$$y_{s,k}(n) - \tilde{R}_{s,k}(n)T_0 \le 0, \quad \forall\, s\in\mathcal{S},\ \forall\, k\in\mathcal{K}, \tag{13f}$$

$$y_{s,k}(n) - M_{s,a_{s,k}(n)}(n) \le 0, \quad \forall\, s\in\mathcal{S},\ \forall\, k\in\mathcal{K}, \tag{13g}$$

$$\lambda_{s,i} \ge 0, \quad \forall\, s\in\mathcal{S},\ \forall\, i\in\Omega(s), \tag{13h}$$

$$\sum_{s\in\Phi(i)} \lambda_{s,i} = \Lambda_i, \quad \forall\, 1\le i\le N_c, \tag{13i}$$

where the index sets of s, k, and l are denoted by $\mathcal{S}$, $\mathcal{K}$, and $\mathcal{L}$, respectively. Moreover, the objective shows that this problem is concerned with long-term performance optimization. Since there is no definite termination state for this sequential decision process, N approaches infinity in the standard form, as discussed in [22], [31], [32], and [33] The above constraints are defined as follows.

- **Coverage constraint**: Constraints (13a) and (13b) denote that each satellite can only serve cells within its coverage area, and the beam of one satellite cannot overlap within the same cell, respectively.
- **Allocation constraint**: $f_{s,k,l}$, being an indicator variable, is restricted to binary values of 0 or 1 as (13c).
- **Power constraint**: Constraints (13d) and (13e) limit the power from exceeding the thresholds of each satellite P^{sat} and the whole satellite network P^{tot}, respectively.
- **Throughput constraint**: The throughput $y_{s,k}(n)$ is ensured to satisfy the definition in (10) by constraints (13f) and (13g).
- **Input traffic constraint**: Constraint (13h) guarantees the non-negativity of arrival rates, while (13i) stipulates that the data rate of input traffic allocated to each satellite in $\Phi(i)$ aligns with the data rate of input traffic Λ_i.

However, solving (P0) directly is challenging. The BH pattern $\boldsymbol{a}_s(n)$ and frequency allocation $\boldsymbol{f}$ are constrained to a discrete feasible region due to constraints (13a), (13b), and (13c), while constraint (13f) is non-linear. Consequently, as a mixed integer non-linear programming problem, the problem (P0) has been proven to be NP-hard [15], [24]. In addition, the feasible space of BH pattern $\boldsymbol{a}_s(n)$ is notably vast, and the current selection also impacts future performance, making it particularly difficult to determine $\boldsymbol{a}_s(n)$. Therefore, it is an intuitive approach and necessary concession to decompose the original problem into three sub-problems on different time scales according to the distinct optimization variables and solve them iteratively, as shown in **Algorithm** 1.

Specifically, the BH pattern $\boldsymbol{a}_s(n)$ is determined by each satellite per time slot in real-time, based on the queue status in its buffer, to handle random traffic arrivals and maximize the long-term rewards. This sub-problem can be further formulated as a Markov decision process (MDP) and solved by DRL approximately. Subsequently, the resource allocation for power P_b and frequency segments $f_{s,k,l}$ is optimized with the given BH pattern in order to mitigate both the intra- and inter-satellite interference in the current time slot. Finally, the load balancing scheme $\boldsymbol{\lambda}$ affects the objective function indirectly and thus can be considered from a long-term perspective as a compromise. Therefore, the load balancing scheme $\boldsymbol{\lambda}$ is replanned at regular intervals T_b, according to the overall network loads in the preceding period. In summary, the overall process is shown in Fig. 2, including DRL for BH scheduling, resource allocation with multi-satellite cooperation, and load balancing via ISLs. Moreover, these sub-problems will be further elucidated in the subsequent sections.

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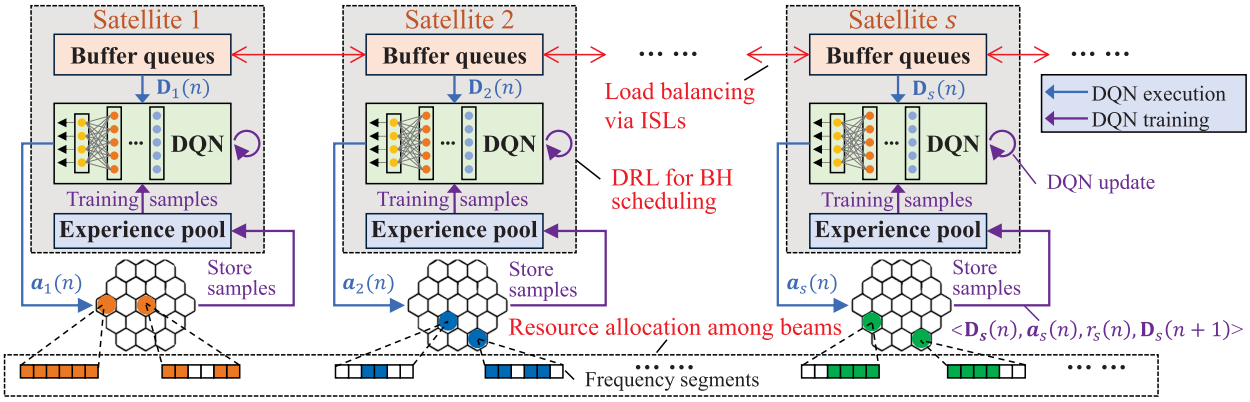

Fig. 2. The proposed multi-satellite cooperative coverage architecture comprised of BH scheduling, resource allocation, and load balancing.

Algorithm 1 The Solution of the Overall Problem

Input: The channel gain $\boldsymbol{h}$, load balacing recalculation interval T_b.

Output: BH pattern series $\boldsymbol{a}_s(n)$, resource allocation results $\boldsymbol{f}$ and P_b, and load balancing scheme $\boldsymbol{\lambda}$

```
Initialize t ← 0.
for Each time slot n do
    Execute the load balancing according to λ.
    Solve BH scheduling sub-problem based on the queue
    status in the buffer of each satellite s, obtain a_s(n).
    Solve resource allocation sub-problem according to BH
    pattern a_s(n), obtain f and P_b.
    t ← t + 1.
    if t = T_b then
        t ← 0, estimate the traffic arrival rate according to
        the preceding period.
        Update the load balancing scheme λ by solving the
        load balancing sub-problem.
    end if
end for
```

IV. Optimization of BH Scheduling

In this section, the first sub-problem concerning BH pattern selection is formulated to maximize the long-term reward. This problem is further transformed into an MDP and solved by multi-agent DRL on each satellite to meet real-time requirements.

A. Problem Formulation for BH

When prerequisites, such as resource allocation rules as well as the load balancing scheme, are specified, the problem (P0) is equivalent to searching a sequence of BH patterns $\boldsymbol{a}_s(n)$ to maximize the following long-term objective as

$$(\text{P1}): \max_{\boldsymbol{a}_s(n)} \lim_{\substack{N\to+\infty \\ \rho\to 1}} \frac{1}{\sum_{n=1}^{N}\rho^{n-1}} \sum_{n=1}^{N}\sum_{s=1}^{N_s}\rho^{n-1} r_s(n), \quad \text{s.t. (13a), and (13b),} \tag{14}$$

where ρ is introduced as the discount factor to convert the average reward into a discounted reward to make the problem solvable [32], and is assigned a constant value less than 1. Moreover, the objective in (P1) tends to focus on the long-term reward when $\rho \to 1$, which is consistent with (P0). Despite the presence of inter-satellite interference, its impact may be alleviated through the allocation of frequency segments. Thereby, the problem (P1) for the whole satellite network can be further decomposed into an optimization problem for each satellite, and the one for satellite s can be formulated as

$$(\text{P1-s}): \max_{\boldsymbol{a}_s(n)} \lim_{N\to+\infty} \sum_{n=1}^{N}\rho^{n-1} r_s(n), \tag{15}$$

$$\text{s.t. } a_{s,k}(n) \in \Omega(s), \quad \forall\, k \in \mathcal{K}, \tag{15a}$$

$$a_{s,j}(n) \neq a_{s,k}(n), \quad \forall\, j,k \in \mathcal{K},\ j \neq k, \tag{15b}$$

in which the constant term $1/\sum_{n=1}^{N}\rho^{n-1}$ is omitted. The objective function of (P1-s) shows that the scheduling of $\boldsymbol{a}_s(n)$ should not only focus on the current time slot, but also consider the long-term reward. Therefore, (P1-s) can be treated as a sequential decision process, which is challenging for conventional optimization methods.

Moreover, when other conditions are fixed and the inter-satellite interference is neglected, $\mathbf{D}_s(n)$ is only related to the BH pattern $\boldsymbol{a}_s(n-1)$ and the state $\mathbf{D}_s(n-1)$ of the previous time slot. Consequently, the sequential decision process (P1-s) can be reformulated as an MDP with the following parameters.

- **State**: The status $\mathbf{D}_s(n)$ of the buffer in satellite s, which can be rewritten as a $|\Omega(s)| \times T_{th}$ matrix.
- **Action**: The BH pattern $\boldsymbol{a}_s(n)$ is set as the action of the MDP, and should meet constraints (13a) and (13b).
- **Reward**: The immediate reward obtained by performing an action in the current state, the reward in the modeled MDP is $r_s(n)$, according to (P1-s).
- **Discount Factor**: The discount rate of the expected forward reward in the current time slot, which is defined as ρ in (P1-s).

Therefore, the optimal $\boldsymbol{a}_s(n)$ selection strategy is sought to achieve the maximum long-term reward in this MDP.

B. DRL for BH Scheduling

To solve the aforementioned MDP, reinforcement learning is a feasible choice. However, the state space is extremely large,

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while the time-varying characteristics of the state are difficult to obtain, and hence a model-free DRL approach is necessary. Therefore, deep Q-network (DQN) is an effective solution, aiming to fit a function $Q(\mathbf{D}_s(n), \boldsymbol{a}_s(n))$, which represents the expected reward that can be obtained after taking action $\boldsymbol{a}_s(n)$ in state $\mathbf{D}_s(n)$. In addition, N_b served cells should be selected from $|\Omega(s)|$ cells, which leads to an extremely large space for $\boldsymbol{a}_s(n)$. To reduce the action space, the adjacent beam avoidance requirement is imposed in [26]. However, this method not only weakens the flexibility of BH, but the action space still increases rapidly as the number of coverage cells grows. In this paper, as an approximation, the DQN outputs the corresponding Q-value function of each cell $\hat{Q}(\mathbf{D}_s(n), i)$, which denotes the contribution to the total Q-value function when the i-th cell is served. Therefore, the total Q-value function can be calculated as

$$Q(\mathbf{D}_s(n), \boldsymbol{a}_s(n); \boldsymbol{\theta}) = \sum_{k=1}^{N_b} \hat{Q}(\mathbf{D}_s(n), a_{s,k}(n); \boldsymbol{\theta}), \tag{16}$$

where $\boldsymbol{\theta}$ denotes the parameters of the DQN. To balance the exploration and exploitation in the DRL, an ϵ-greedy strategy is adopted to determine action $\boldsymbol{a}_s(n)$ chosen by agent i as

$$\boldsymbol{a}_s(n) = \begin{cases} \arg\max_{\boldsymbol{a}} Q(\mathbf{D}_s(n), \boldsymbol{a}; \boldsymbol{\theta}), & \text{with probability 1-}\epsilon, \\ \text{random selection}, & \text{with probability } \epsilon, \end{cases} \tag{17}$$

where ϵ denotes the probability of an exploration by the agent (i.e., satellite s). After the agent takes action, it goes to the next state $\mathbf{D}_s(n{+}1)$ and gets a reward $r_s(n)$, then the resulting experience sample $< \mathbf{D}_s(n), \boldsymbol{a}_s(n), r_s(n), \mathbf{D}_s(n{+}1) >$ is stored in the experience pool $\mathcal{E}_s$. By learning from the samples in the experience pool, DQN can effectively infer the transition of the state. Consequently, it does not need to know the data rate for each time slot in advance. In addition, to avoid overestimation of the Q-value function, double DQN utilizes two DQNs with identical architecture, including the current network with parameters $\boldsymbol{\theta}$ and the target network with parameters $\boldsymbol{\theta}^-$ [34]. During training, the samples are chosen from the experience pool to calculate the target Q-value function according to the Bellman equation [35] as

$$\begin{aligned} &Q'(\mathbf{D}_s(n), \boldsymbol{a}_s(n)) \\ &= r_s(n) + \rho Q\big(\mathbf{D}_s(n{+}1), \arg\max_{\boldsymbol{a}} Q(\mathbf{D}_s(n{+}1), \boldsymbol{a}; \boldsymbol{\theta}); \boldsymbol{\theta}^-\big). \end{aligned} \tag{18}$$

Therefore, the loss function is expressed as

$$L(\boldsymbol{\theta}) = \left|Q'(\mathbf{D}_s(n), \boldsymbol{a}_s(n)) - Q(\mathbf{D}_s(n), \boldsymbol{a}_s(n); \boldsymbol{\theta})\right|^2, \tag{19}$$

and the parameters of the current network $\boldsymbol{\theta}$ are updated according to the loss function, while those of the target network $\boldsymbol{\theta}^-$ are replaced by $\boldsymbol{\theta}$ in every C_u time slots. In summary, the overall process of the DRL for the BH scheduling can be found in **Algorithm** 2.

The computational complexity is primarily attributed to the training and inference of DQN in steps 10∼12 and steps 14∼15. Moreover, the full-connected layer is utilized to compose the DQN, while Z_m^{in} and Z_m^{out} denote the input and output size of the m-th layer. Therefore, the computational complexity of **Algorithm** 2, which is proportional to the number of parameters in DQN, can be quantifiable as $\mathcal{O}(\sum_m Z_m^{\text{in}} \times Z_m^{\text{out}})$, respectively. Moreover, despite the extensive computational demands required during the training phase, there are numerous parallelizable operations [36].

Algorithm 2 The DRL for the BH Scheduling

Input: The number of time slots Θ_2.
Output: The BH pattern series $\boldsymbol{a}_s(n)$.

1: **for** LEO satellite $s = 1$ to N_s **do**
2: Initialize $\mathbf{D}_s(0)$ to an all-zero matrix, and $r_s(0) = 0$.
3: Initialize $\boldsymbol{a}_s(n)$ by random selection from $\Omega(s)$.
4: **end for**
5: **for** time slot $n = 1$ to Θ_2 **do**
6: Execute load balancing and arrival of data packets.
7: **for** LEO satellite $s = 1$ to N_s **do**
8: Satellite s updates its buffer state $\mathbf{D}_s(n)$.
9: Store the experience sample $< \mathbf{D}_s(n-1), \boldsymbol{a}_s(n-1), r_s(n-1), \mathbf{D}_s(n) >$ into the experience pool $\mathcal{E}_s$.
10: Collect the samples from $\mathcal{E}_s$.
11: Calculate the loss function by (19).
12: Update the current network to $\boldsymbol{\theta}$.
13: Update the target network with $\boldsymbol{\theta}^- \leftarrow \boldsymbol{\theta}$ every C_u time slots.
14: Input $\mathbf{D}_s(n)$ to the current network, calculate the Q-value function of each cell $\hat{Q}(\mathbf{D}_s(n), i; \boldsymbol{\theta})$.
15: Choose the BH pattern $\boldsymbol{a}_s(n)$ with the ϵ-greedy strategy according to (16) and (17).
16: **end for**
17: Calculate the resource allocation, and transmission data.
18: **for** LEO satellite $i = 0$ to N_s **do**
19: Calculate the reward $r_s(n)$ according to (12).
20: **end for**
21: **end for**

V. Resource Allocation Among Beams

After determining the BH pattern of each satellite, the sub-problem associated with resource allocation is optimized to mitigate interference in this section, thereby improving the throughput of the satellite network. Specifically, the MM algorithm converts the sub-problem into a sequence of convex optimization problems, which are solved iteratively. Finally, the computational complexity and overhead of the resource allocation are analyzed.

A. Problem Formulation for Interference Mitigation

Given the BH pattern of the current time slot determined by the aforementioned DRL, the frequency segments and power can be further allocated to mitigate interference, thereby enhancing the throughput of the satellite network. Hence, the

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optimization problem can be expressed as

$$\text{(P2)}: \max_{P_b, \boldsymbol{f}} \sum_{s=1}^{N_s}\sum_{k=1}^{N_b} y_{s,k}, \quad \text{s.t. (13c), (13d), (13e), (13f), and (13g).} \tag{20}$$

Since the frequency segments are allocated solely for the immediate time slot, the part of the variable name adopted to represent the current time slot is omitted in (P2) for convenience.

However, since $f_{s,k,l}$ is a discrete variable, the resolution of (P2) constitutes an NP-hard problem. To provide an efficient, albeit suboptimal solution, constraint (13c) can be relaxed as

$$0 \le f_{s,k,l} \le 1. \tag{21}$$

Moreover, an auxiliary variable $z = 1/P_b$ is introduced to linearize constraints (13d) and (13e) as

$$\sum_{s=1}^{N_s}\sum_{k=1}^{N_b}\sum_{l=1}^{N_L} f_{s,k,l} \le P^{tot} z, \tag{22}$$

$$\sum_{k=1}^{N_b}\sum_{l=1}^{N_L} f_{s,k,l} \le P^{sat} z, \quad \forall\, s \in \mathcal{S}, \tag{23}$$

and thus (P2) is relaxed as

$$\text{(P2-1)}: \max_{P_b, \boldsymbol{f}} \sum_{s=1}^{N_s}\sum_{k=1}^{N_b} y_{s,k}, \quad \text{s.t. (13f), (13g), (21), (22), and (23).} \tag{24}$$

Upon determining the solution $\boldsymbol{f}^\star$ and $P_b^\star$ of (P2-1), P_b is initially set to $P_b^\star$, then the sequence $\boldsymbol{f}$ is sorted based on the descending order of $\boldsymbol{f}^\star$ and sequentially assigned to ones until constraints (13c) and (13d) are exceeded, remaining elements are assigned to zeros. Subsequently, P_b is decremented to exactly satisfy both constraints (13c) and (13d).

B. Majorization-Minimization for Resource Allocation

To solve the problem of resource allocation, it must be ascertained whether (P2-1) exhibits convexity. Since other constraints have been converted into linear constraints, we have the following proposition.

Proposition 1: $y_{s,k} - \tilde{R}_{s,k} T_0$ is not a convex function.

Proof: See Appendix A.

Consequently, the feasible region of (P2-1) is non-convex due to the constraint (13f), while the MM algorithm offers a viable approach to address such issues. Through the identification of a convex surrogate function, the feasible region is approximated as convex, allowing for iterative convergence to the Karush-Kuhn-Tucker (KKT) point of (P2-1). In the q-th iteration, the initial values of $\boldsymbol{f}$ and P_b are set to $\boldsymbol{f}^{(q)}$ and $P_b^{(q)}$, respectively. Meanwhile, for convenience, the following terms are defined as

$$\gamma_{s,k,l}^{(q)} = \gamma_{s,k,l}(\boldsymbol{f}^{(q)}, P_b^{(q)}), \tag{25a}$$

$$\phi_{s,k,l}^{(q)} = \frac{\gamma_{s,k,l}^{(q)}}{1+\gamma_{s,k,l}^{(q)}}, \tag{25b}$$

$$\zeta_{s,k,l}^{(q)} = \log_2(1+\gamma_{s,k,l}^{(q)}) - \phi_{s,k,l}^{(q)} \log_2(\gamma_{s,k,l}^{(q)}), \tag{25c}$$

$$\alpha_{s,k} = \frac{n_0 B_0}{\left|h_{s,k}^{s,k}\right|^2}, \tag{25d}$$

and thus a surrogate function $\underline{R}_{s,k,l}^{(q)}$ can be calculated as

$$\underline{R}_{s,k,l}^{(q)} = B_0\Big(\big(\zeta_{s,k,l}^{(q)} + \phi_{s,k,l}^{(q)}\log_2(\gamma_{s,k,l}^{(q)} e)\big) f_{s,k,l} - \phi_{s,k,l}^{(q)}\gamma_{s,k,l}^{(q)}\log_2(e)\frac{f_{s,k,l}}{\gamma_{s,k,l}(\boldsymbol{f}, P_b)}\Big). \tag{26}$$

Moreover, $\underline{R}_{s,k,l}^{(q)}$ is a linear function due to the following expression as

$$\frac{f_{s,k,l}}{\gamma_{s,k,l}(\boldsymbol{f}, P_b)} = z\alpha_{s,k} + \sum_{\substack{r=1,\\ r\neq s}}^{N_s}\sum_{j=1}^{N_b} f_{r,j,l}\frac{\left|h_{s,k}^{r,j}\right|^2}{\left|h_{s,k}^{s,k}\right|^2} + \sum_{\substack{j=1,\\ j\neq k}}^{N_b} f_{s,j,l}\frac{\left|h_{s,k}^{s,j}\right|^2}{\left|h_{s,k}^{s,k}\right|^2}, \tag{27}$$

and the properties of $\underline{R}_{s,k,l}^{(q)}$ are described in the following proposition.

Proposition 2: The surrogate function $\underline{R}_{s,k,l}^{(q)}$ has the following properties

$$R_{s,k,l} \ge \underline{R}_{s,k,l}^{(q)}, \tag{28}$$

$$R_{s,k,l}\big|_{\boldsymbol{f}=\boldsymbol{f}^{(q)}, P_b=P_b^{(q)}} = \underline{R}_{s,k,l}^{(q)}\big|_{\boldsymbol{f}=\boldsymbol{f}^{(q)}, P_b=P_b^{(q)}}, \tag{29}$$

$$\nabla R_{s,k,l}\big|_{\boldsymbol{f}=\boldsymbol{f}^{(q)}, P_b=P_b^{(q)}} = \nabla \underline{R}_{s,k,l}^{(q)}\big|_{\boldsymbol{f}=\boldsymbol{f}^{(q)}, P_b=P_b^{(q)}}. \tag{30}$$

Proof: See Appendix B.

Therefore, constraint (13f) can be transformed into a linear constraint as

$$y_{s,k} - \sum_{l=1}^{N_L} \underline{R}_{s,k,l}^{(q)} T_0 \le 0, \ \forall\, s \in \mathcal{S}, \ \forall\, k \in \mathcal{K}. \tag{31}$$

Therefore, the KKT solution of (P2-1) can be obtained by solving a series of approximate linear programming problems iteratively as

$$\text{(P2-2)}: \max_{P_b, \boldsymbol{f}} \sum_{s=1}^{N_s}\sum_{k=1}^{N_b} y_{s,k}, \quad \text{s.t. (13g), (21), (22), (23), and (31).} \tag{32}$$

At the outset of the q-th iteration, the resultant values $\boldsymbol{f}^{(q)}$ and $P_b^{(q)}$ from the preceding iteration are employed to calculate constraint (31). Subsequently, the linear programming problem (P2-2) is resolved to yield $\boldsymbol{f}^{(q+1)}$ and $P_b^{(q+1)}$, thus initiating the $(q+1)$-th iteration. Finally, the sequence generated through this iterative process will converge to the KKT point of (P2-1) as proven in [37].

C. Complexity and Overhead Analysis

According to the aforementioned method, the solution of (P2) is shown in **Algorithm** 3, and the solution of the relaxed (P2-1) is obtained by iteratively solving the approximate linear programming problem (P2-2) obtained by the MM algorithm, as shown in steps 1∼5. Then, P_b and $\boldsymbol{f}$ are adjusted to conform to the constraints of (P2) by heuristic

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Algorithm 3 Resource Allocation Based on MM Algorithm

Input: The BH pattern $\boldsymbol{a}_s$, channel gain $\boldsymbol{h}$.
Output: Solutions $\boldsymbol{f}$, P_b.

1: Initialize a feasible solution $\boldsymbol{f}^{(1)}$, $P_b^{(1)}$, and $q \leftarrow 1$.
2: **repeat**
3: Update the problem (P2-2) with $\boldsymbol{f}^{(q)}$ and $P_b^{(q)}$.
4: Solve the linear programming problem (P2-2) and obtain $\boldsymbol{f}^{(q+1)}$, $P_b^{(q+1)}$.
5: $q \leftarrow q+1$
6: **until** $\boldsymbol{f}^{(q)}$ and $P_b^{(q)}$ converge
7: Fixed $P_b^{(q)}$, sort $\boldsymbol{f}$ according the descending order of $\boldsymbol{f}^{(q)}$.
8: Assign the elements of $\boldsymbol{f}$ to ones sequentially until the constraints (13c) and (13d) are exceeded, and set remaining part to zeros.
9: Adjust P_b to satisfy both constraints (13c) and (13d).

methods in steps 6∼8. For the computational complexity, solutions to a series of linear programming problems are required, which can be efficiently solved in polynomial time by the interior point method with a complexity of $\mathcal{O}\big((N_s N_b N_L)^{3.5}\log(1/\xi_3)\big)$, where ξ_3 represents the specified solution accuracy [38]. Additionally, the MM algorithm exhibits linear convergence [39], resulting in a total number of iterations denoted by $\mathcal{O}\big(\log(1/\xi_3)\big)$. Consequently, the overall computational complexity of **Algorithm** 3 is $\mathcal{O}\big((N_s N_b N_L)^{3.5}(\log(1/\xi_3))^2\big)$.

For the cooperation overhead, the BH pattern and queue status of each satellite need to be obtained, which have little impact on bandwidth and energy consumption due to the small data size. However, the extensive distances between satellites cause significant delays in information exchange, with the round-trip delay for one hop of the ISL being on the order of milliseconds [27], making it a major bottleneck in multi-satellite cooperation. Therefore, as the number of hops required for inter-satellite cooperation increases, the time overhead for satellite cooperation becomes untenable. A practical solution is to limit the scope of multi-satellite cooperation such that each satellite only cooperates and optimizes with its neighboring satellites separately. Moreover, simulation results in Section VII also indicate that this strategy can reduce the cooperative overhead with slight performance loss.

VI. Load Balancing Among Satellites

The load balancing scheme $\boldsymbol{\lambda}$ implicitly affects the objective function, and thus it should be considered from a long-term perspective. In this section, the sub-problems for load balancing in different scenarios are formulated, and then solved by linear programming or hybrid block successive approximation for various objectives.

A. Load Balancing in Different Scenarios

The primary goal of load balancing is to minimize the average delay of the input traffic when the achievable data rate of the satellite network is sufficient to fulfill the input traffic. Conversely, when the input traffic surpasses the serviceability of the satellite network, the aim is to ensure an equitable load distribution among satellites. The service ability of each satellite can be estimated by averaging the throughput over a period of time, and the serviceability of the beam from satellite s to the i-th cell is denoted as $U_{s,i}$.

When the network serviceability is sufficient to handle all arrival traffic, the average service rate allocated to the i-th cell by satellite s is denoted as $\mu_{s,i}$, while the vector composed of all $\mu_{s,i}$ is denoted by $\boldsymbol{\mu}$. Thus, the beams of satellite s serve the i-th cell every $U_{s,i}/\mu_{s,i}$ time slots on average. Moreover, the queue could be cleared within the time slot during the service, to minimize the average delay of data packets. In this context, the average waiting time of the queue $\boldsymbol{d}_{s,i}(n)$ can be calculated as

$$t_{s,i} = \frac{1}{2}\frac{U_{s,i}}{\mu_{s,i}}. \tag{33}$$

Therefore, considering the additional delay caused by load balancing, the optimization problem can be formulated as

$$\text{(P3a)}: \min_{\boldsymbol{\lambda},\boldsymbol{\mu}} \sum_{s=1}^{N_s}\sum_{i\in\Omega(s)} \lambda_{s,i} t_{s,i} + \sum_{i=1}^{N_c} \Delta\lambda_i, \tag{34}$$

$$\text{s.t. (13h), and (13i)},$$

$$\sum_{i\in\Omega(s)} \frac{\mu_{s,i}}{U_{s,i}} \le N_b, \quad \forall\, s\in\mathcal{S}, \tag{34a}$$

$$\mu_{s,i} > \lambda_{s,i}, \quad \forall\, s\in\mathcal{S},\ \forall\, 1\le i\le N_c, \tag{34b}$$

in which the constraint of (34a) indicates that the sum of the service rate provided for each cell cannot exceed the service ability of the corresponding satellite, while (34b) represents that the service rate should larger than the arrival rate. Moreover, although the solution of (P3a) provides the distribution of the service rate $\boldsymbol{\mu}$, representing a macroscopic perspective, the BH pattern specific to each time slot still needs to be selected through the DRL as mentioned in Section IV.

When constraints (34a) and (34b) cannot be satisfied, which means the input traffic surpasses the service ability of the network, there is no feasible solution for (P3a). The packet loss and high latency are unavoidable in this scenario. Therefore, the aim of load balancing at this time is to minimize the difference in load among satellites, which is given by

$$\text{(P3b)}: \min_{\boldsymbol{\lambda}} \overline{U} - \underline{U}, \tag{35}$$

$$\text{s.t. (13h), and (13i)},$$

$$\sum_{i\in\Omega(s)} \frac{\lambda_{s,i}}{U_{s,i}} \le \overline{U}, \quad \forall\, s\in\mathcal{S}, \tag{35a}$$

$$\sum_{i\in\Omega(s)} \frac{\lambda_{s,i}}{U_{s,i}} \ge \underline{U}, \quad \forall\, s\in\mathcal{S}, \tag{35b}$$

where $\overline{U}$ in (35a) and $\underline{U}$ in (35b) represent the maximum and minimum traffic loads among satellites, respectively, while the problem (P3b) is a linear programming problem that can be easily solved with a toolbox such as GUROBI. Therefore, the subsequent discussion exclusively focuses on the solution for the problem (P3a).

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B. Successive Approximation Algorithm for Load Balancing

For the feasible region of (P3a), it is an open set due to (34b). For convenience, choose a $\delta \rightarrow 0^+$, and constraint (34b) can be rewritten as

$$\mu_{s,i} \geq \lambda_{s,i} + \delta, \quad \forall\, s \in \mathcal{S},\ \forall\, 1 \leq i \leq N_c. \tag{36}$$

Moreover, the objective of (P3a) comprises a sum of absolute terms $\sum_{i=1}^{N_c} \Delta\lambda_i$ and a sum of fractional terms $\sum_{s=1}^{N_s}\sum_{i\in\Omega(s)} \lambda_{s,i} t_{s,i}$. The former can be solved by variable substitution as

$$v_{s,i} = |\lambda_{s,i} - \lambda_i^-|, \tag{37}$$

and then the following linear constraints are added to ensure the equivalence as

$$v_{s,i} \geq \lambda_{s,i} - \lambda_i^-, \tag{38}$$

$$v_{s,i} \geq \lambda_i^- - \lambda_{s,i}. \tag{39}$$

Therefore, the sum of absolute terms can be addressed as linear objectives. For the fractional terms in the latter, the quadratic transformation [40] is employed to reformulate (P3a) into the subsequent problem as

$$\begin{aligned}(\text{P3a-1}):\ & \min_{\boldsymbol{\lambda},\boldsymbol{\mu}} \max_{\boldsymbol{g}} \sum_{s=1}^{N_s}\sum_{i\in\Omega(s)} G_{s,i} + \frac{1}{2}\sum_{i=1}^{N_c}\sum_{s\in\Phi(i)} v_{s,i},\\ & \text{s.t. (13h), (13i), (34a), (36), (38), and (39),}\end{aligned} \tag{40}$$

where $G_{s,i}$ denotes a quadratic transformation of the fraction term with an auxiliary variable $g_{s,i}$, and is given by

$$G_{s,i} = \frac{U_{s,i}}{2}\left(2g_{s,i}\sqrt{\lambda_{s,i}} - g_{s,i}^2\mu_{s,i}\right). \tag{41}$$

Moreover, $\boldsymbol{g}$ represents the set of all $g_{s,i}$. Therefore, the convexity of $G_{s,i}$ needs to be considered. Since $G_{s,i}$ is strongly concave in $g_{s,i}$, caused by (13h) and (36), but not convex in $\lambda_{s,i}$, the problem (P3a-1) is a one-sided non-convex min-max problem, whose KKT solution can be obtained by hybrid block successive approximation algorithm in an iterative manner [41].

The specific detail of iteration is that, after completing the $(p-1)$-th iteration, the parameters $\boldsymbol{\lambda}^{(p)}$, $\boldsymbol{\mu}^{(p)}$, and $\boldsymbol{g}^{(p)}$ are obtained, which can initiate the p-th iteration. With $\boldsymbol{g}^{(p)}$ held constant, the minimization problem involving $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$ is to be addressed first. Given that $G_{s,i}$ is a non-convex function concerning $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$, the successive convex approximation algorithm is further employed, and a convex upper bound $\overline{G}_{s,i}^{(p)}$ is introduced as

$$\begin{aligned}\overline{G}_{s,i}^{(p)} = & \frac{U_{s,i}}{2}\Big(\Big(\sqrt{\lambda_{s,i}^{(p)}} + \frac{1}{\sqrt{\lambda_{s,i}^{(p)}}}\lambda_{s,i}\Big)g_{s,i}^{(p)} - \mu_{s,i}\big(g_{s,i}^{(p)}\big)^2\Big)\\ & + \eta\big((\lambda_{s,i} - \lambda_{s,i}^{(p)})^2 + (\mu_{s,i} - \mu_{s,i}^{(p)})^2\big),\end{aligned} \tag{42}$$

where η is a parameter to control the stability and convergence speed. Moreover, when $\boldsymbol{g}^{(p)}$ is fixed, it is easy to prove that

$$\overline{G}_{s,i}^{(p)} \geq G_{s,i}, \tag{43}$$

$$\overline{G}_{s,i}^{(p)}|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}^{(p)},\ \boldsymbol{\mu}=\boldsymbol{\mu}^{(p)}} = G_{s,i}|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}^{(p)},\ \boldsymbol{\mu}=\boldsymbol{\mu}^{(p)}}, \tag{44}$$

$$\nabla_{\boldsymbol{\lambda},\boldsymbol{\mu}}\overline{G}_{s,i}^{(p)}|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}^{(p)},\ \boldsymbol{\mu}=\boldsymbol{\mu}^{(p)}} = \nabla_{\boldsymbol{\lambda},\boldsymbol{\mu}} G_{s,i}|_{\boldsymbol{\lambda}=\boldsymbol{\lambda}^{(p)},\ \boldsymbol{\mu}=\boldsymbol{\mu}^{(p)}}. \tag{45}$$

Algorithm 4 Load Balancing by Hybrid Block Successive Approximation

Input: The service capacity $U_{s,i}$ and packets arrival rate Λ_i, estimated by last T_b time slots.
Output: Load balancing strategy $\boldsymbol{\lambda}$ for next T_b time slots.

1: **if** (P3a) has a feasible solution, **then**
2: Initialize a feasible solution $\boldsymbol{\lambda}^{(1)}$ and $\boldsymbol{\mu}^{(1)}$ of (P3a), calculate $\boldsymbol{g}^{(1)}$ as (47), and $p \leftarrow 1$.
3: **repeat**
4: Solve (P3a-2), and obtain $\boldsymbol{\lambda}^{(p+1)}$ and $\boldsymbol{\mu}^{(p+1)}$.
5: Update $\boldsymbol{g}^{(p+1)}$ according to (47).
6: $p \leftarrow p+1$
7: **until** $\boldsymbol{\lambda}^{(p)}$ and $\boldsymbol{\mu}^{(p)}$ converge
8: $\boldsymbol{\lambda} \leftarrow \boldsymbol{\lambda}^{(p)}$.
9: **else**
10: Solve the linear programming problem (P3b).
11: **end if**

Therefore, $\boldsymbol{\lambda}^{(p+1)}$ and $\boldsymbol{\mu}^{(p+1)}$ can be acquired by solving the optimization problem as

$$\begin{aligned}(\text{P3a-2}):\ & \min_{\boldsymbol{\lambda},\boldsymbol{\mu}} \sum_{s=1}^{N_s}\sum_{i\in\Omega(s)} \overline{G}_{s,i}^{(p)} + \frac{1}{2}\sum_{i=1}^{N_c}\sum_{s\in\Phi(i)} v_{s,i},\\ & \text{s.t. (13h), (13i), (34a), (36), (38), and (39),}\end{aligned} \tag{46}$$

which is a quadratic convex programming that can be solved easily, thereby obtaining $\boldsymbol{\lambda}^{(p+1)}$ and $\boldsymbol{\mu}^{(p+1)}$. Then, fix $\boldsymbol{\lambda}^{(p+1)}$ and $\boldsymbol{\mu}^{(p+1)}$ to solve the maximization problem with respect to $\boldsymbol{g}$, which can be calculated as

$$g_{s,i}^{(p+1)} = \frac{\sqrt{\lambda_{s,i}^{(p+1)}}}{\mu_{s,i}^{(p+1)}}, \tag{47}$$

and then the $(p+1)$-th iteration can be started until convergence.

In summary, the solution of load balancing is demonstrated in **Algorithm** 4. If the arrival rate does not exceed the overall service ability of the satellite network, the hybrid block successive approximation method is adopted to iteratively solve the (P3a-1) after equivalent transformation from (P3a), to reduce the average waiting time, as shown in steps 1∼7. Otherwise, (P3b) is solved to balance the load among satellites. In the context of computational complexity, solving (P3a) requires the iterative solution of a sequence of approximate convex optimization problems (P3a-2) via the block successive approximation method, as outlined in steps 3∼6, with a complexity of $\mathcal{O}\big((N_s|\Omega(s)|)^{3.5}\log(1/\xi_4)\big)$. Moreover, drawing from the conclusion in [41], the total iteration count is $\mathcal{O}\big(1/(\xi_4)^2\big)$. Consequently, the complexity of addressing (P3a) is quantified as $\mathcal{O}\big((N_s|\Omega(s)|)^{3.5}\log(1/\xi_4)/(\xi_4)^2\big)$. Conversely, addressing (P3b) necessitates solving only the linear program in step 9. As mentioned above, given a solution accuracy of ξ_4, the complexity is $\mathcal{O}\big((N_s|\Omega(s)|)^{3.5}\log(1/\xi_4)\big)$ by the interior point method.

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TABLE I
SIMULATION PARAMETERS

PARAMETERS	VALUE
Satellite-Terrestrial Communications Parameters	
The satellite altitude	1000 km
The number of cells N_c	382
The number of cells for each satellite $\|\Omega(s)\|$	37
The number of frequency segments N_L	10
The Ka-band frequency	20 GHz
The bandwidth of each frequency segment B_0	10 MHz
The number of beam in one satellite N_b	8
The radius of each cell	38 km
The noise spectral power density n_0	-204 dBW
The power limitation of each satellite P^{sat}	40.00 dBW
The power limitation of total network P^{tot}	52.04 dBW
The transmit antenna gain	32.44 dBi
The receive antenna gain	0 dBi
The buffer reservation threshold T_{th}	40
The coefficient κ in latency metric	0.1
The data arrival rate Λ_i	35∼105 Mbps
The time slot duration T_0	10 ms
The normalization constants Υ_0, Γ_0	0.5 Mbit, 1
The load balancing recalculation interval T_b	2000
The DRL Parameters	
The discount factor ρ	0.99
The learning rate	0.0001
The exploration probability ϵ	0∼1
The target network update frequency C_u	1000

TABLE II
STRUCTURE OF DQN

Layer	**Input** Z_m^{in}	**Output** Z_m^{out}	**Activation**
1	$\|\Omega(s)\| \times T_{th}$	2048	Relu
2	2048	1024	Relu
3	1024	512	Relu
4	512	256	Relu
5	256	128	Relu
6	128	$\|\Omega(s)\|$	Linear

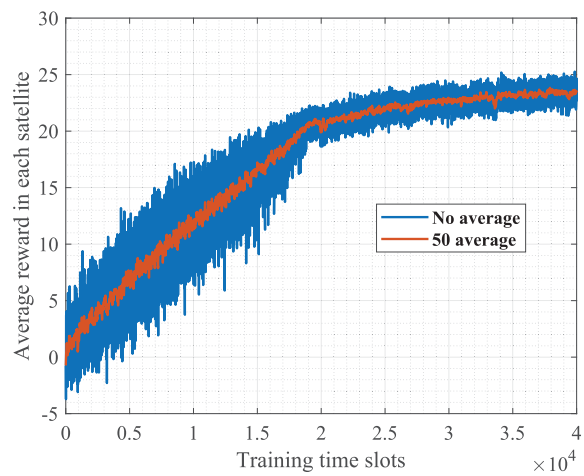

Fig. 3. The average reward in each satellite during the training period with $\beta = 0.7$, and the total data rate of input traffic is 35 Gbps.

VII. SIMULATION AND PERFORMANCE ANALYSIS

In this section, the convergence of the algorithm is analyzed and simulated. The parameters utilized in the simulation are presented in Table I. Specifically, the satellite orbit altitude is set to 1000 km, as the altitudes of the several LEO satellite networks are 550∼1200 km [5]. Moreover, the transmission frequency is set to the Ka-band at 20 GHz, while a circular antenna with an aperture radius of 0.1 m is utilized, and the cell radius is determined based on the 3 dB beamwidth of the antenna gain [24]. Furthermore, the duration of each time slot is chosen to be 10 ms according to [12]. Additionally, in order to simulate the non-uniform geographical and time-varying distribution of traffic, as well as the motion of the satellites, the initial data rate of input traffic for each cell is randomly distributed between 35 ∼105 Mbps. Meanwhile, every 200 time slots, the data rate of each cell undergoes a random walk within a range of ±10 Mbps from its current value, while ensuring it remains within the specified bounds. Moreover, the structure of the DQN, which is composed of six full-connected layers, is shown in Table II, and the DQNs on each satellite share identical parameters. Meanwhile, the effectiveness of the proposed method is verified by comparison with other benchmarks in various simulation scenarios. The simulation results for different methods are obtained by averaging the outcomes over 2,000 time slots under identical parameters.

A. Convergence Analysis and Simulation

It has been proven in [42] that DQN with ϵ-greedy exploration strategy, which is utilized in **Algorithm** 2, converges geometrically. Moreover, the rewards for each time slot during the DQN training period for the optimization of the BH pattern are illustrated in Fig. 3, with $\beta = 0.7$ and the total data rate of input traffic of 35 Gbps. Meanwhile, to facilitate a clearer observation of the trend, the sliding average results across 50 time slots are also plotted. It is observed that the reward gradually improves and ultimately stabilizes, thereby verifying the convergence of the proposed DRL method in **Algorithm** 2.

Moreover, for the convergence of **Algorithm** 3, the MM algorithm in steps 1∼5 performs linear convergence. Therefore, after $\mathcal{O}(\log(\xi_3))$ iterations, the KKT solution of (P2-1) is obtained with solution accuracy ξ_3. Then, the solutions of (P2) can be achieved with steps 6∼8. Furthermore, the convergence performances of whole constellation optimization, cooperation with neighboring satellites, and no cooperation are shown in Fig. 4. It can be observed that all curves converge quickly. Meanwhile, compared to whole constellation optimization, the performance without cooperation is less effective due to the lack of inter-satellite interference mitigation. In contrast, the performance loss of cooperation with neighboring satellites is less than 0.8%, while the computational complexity and extra overhead do not increase with the constellation expansion. Therefore, cooperation with neighboring satellites is adopted in the following simulations.

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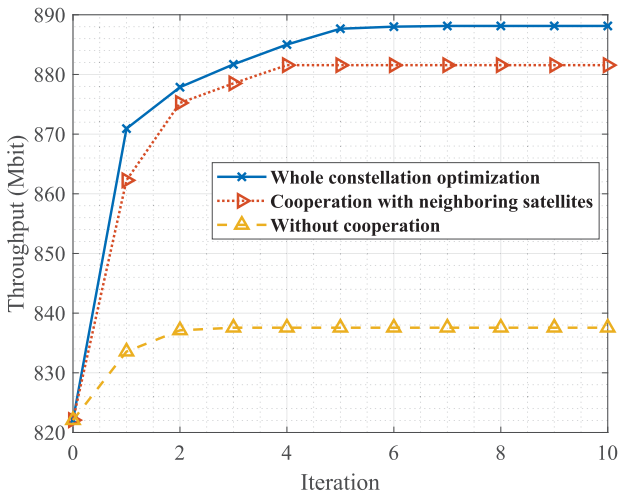

Fig. 4. The convergence performance of **Algorithm** 3 in whole constellation optimization, cooperation with neighboring satellites, and no cooperation.

For **Algorithm** 4, the optimization problem (P3a) is equivalently transformed into a one-sided non-convex min-max problem (P3a-1), which is iteratively solved in steps 3∼6 by hybrid block successive approximation algorithm. Moreover, the KKT solution of (P3a-1) can be obtained after $\mathcal{O}\big(1/(\xi_4)^2\big)$ iterations with solution accuracy ξ_4 [41], thereby obtain the solution of (P3a). Meanwhile, the linear programming problem (P3b) can be solved directly.

B. Comparison With Other Methods

In this subsection, the proposed method is compared with the multi-satellite DRL approach presented in [26], where the adjacent beam avoidance requirement is imposed to prevent inter-beam interference and reduce the action space, thereby eliminating the need for resource allocation. Additionally, the maximum USWG method [21] is evaluated, which considers both the queue length and queuing delay to determine the priority of each cell. Besides, the pre-scheduling BH time plan method [14] is also compared. Moreover, although the adjacent beam avoidance requirement in [26] significantly reduces the action space, the feasible number of action selections remains substantial and challenging to implement when $|\Omega(s)| = 37$. Therefore, this comparison employs $|\Omega(s)| = 19$ cells per satellite to cover $N_c = 160$ cells with $N_s = 16$ satellites, while each satellite is equipped with $N_b = 4$ beams, and the trade-off coefficient $\beta = 0.7$.

The results on throughput are shown in Fig. 5(a). It can be observed that the proposed method outperforms other benchmarks. The pre-scheduling method [14] is less effective, as the BH pattern for each time slot is determined by a pre-planned BH time plan, rendering it less adaptable to time-varying traffic demand. The method incorporating adjacent beam avoidance to reduce interference [26] is inferior to the proposed method, as this constraint reduces the flexibility of BH, leading to suboptimal performance in large areas with high load, as demonstrated in [15]. Moreover, the approach in [26] tends to cause satellite beams to cover unserved cells, and it directly deactivates these beams, which can result in the underutilization of beam resources and thus decrease the performance. Besides, the maximum USWG method [21] making real-time decisions based on the current state, exhibits better performance than the pre-scheduling method. However, it does not consider long-term performance, causing less effective outcomes compared to the other two DRL-based methods.

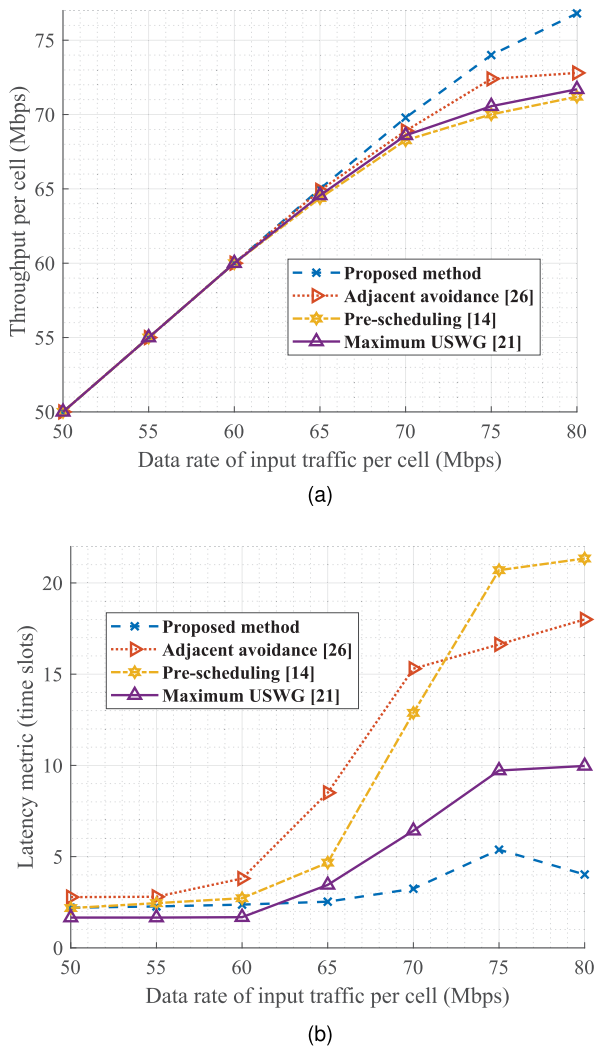

Fig. 5. The performance of different methods with various data rates of input traffic versus (a) the throughput per cell and (b) the latency metric.

For the latency metric shown in Fig. 5(b), the pre-scheduling method [14] does not account for this factor when planning the BH pattern, leading to a noticeable decline in performance under high load conditions. The method with adjacent beam avoidance [26] also experiences challenges with latency metric under high load, due to its constrained BH flexibility, which limits its ability to achieve delay fairness among cells. Additionally, the maximum USWG strategy [21] exhibits the best latency metric in low load conditions, as under low load, the queue latency becomes the primary factor to consider, thereby effectively reducing the overall latency metric. Moreover, the proposed method maintains stable performance for the latency metric across all load levels and achieves the best results under high load conditions.

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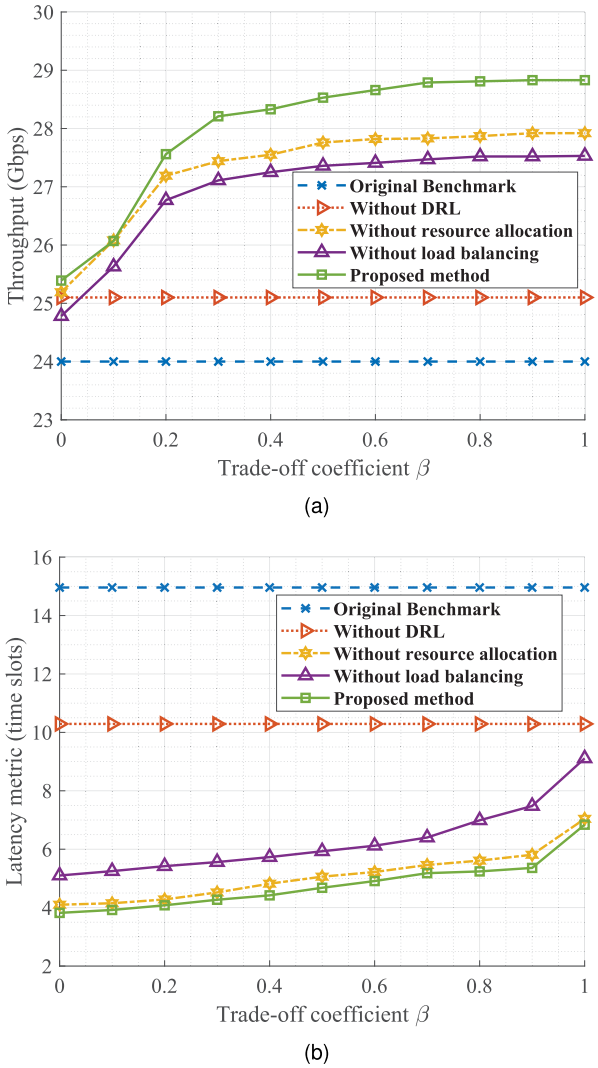

Fig. 6. The performance of different trade-off coefficients β versus (a) the throughput and (b) the latency metric, with the data rate of input traffic of 35 Gbps.

C. Simulation Results in Different Scenarios

In addition to the proposed method, comparisons are performed with the other four benchmarks in this section, including (a) **Original Benchmark**: The BH scheduling is based on a greedy algorithm, whereby the queue holding the largest data volume is prioritized for transmission, neither performing resource allocation among beams nor load balancing via ISLs. (b) **Without DRL**: The greedy algorithm is utilized for BH scheduling, instead of the DRL in the proposed method. (c) **Without Resource Allocation**: This approach excludes resource allocation among beams from the proposed method. (d) **Without Load Balancing**: In this method, load balancing via ISLs is omitted from the proposed method.

1) Results for Different Trade-off Coefficient β: The throughput of satellite-terrestrial communications and the latency metric with different trade-off coefficient β are shown in Fig. 6, where the data rate of input traffic is set at 35 Gbps and $N_s = 16$. It can be observed from Fig. 6(a) that the proposed method outperforms other benchmarks in terms of the throughput. By respectively comparing the proposed method with the corresponding curves of benchmarks without DRL, resource allocation, and load balancing, the simulation results show that the BH scheduling based on DRL, resource allocation among beams, and load balancing via ISLs significantly enhance the throughput of satellite-terrestrial communications. Notably, the impact of the DRL for BH scheduling is the most pronounced.

In the context of the latency metric in Fig. 6(b), both BH scheduling based on DRL and load balancing via ISLs substantially reduce the latency metric in satellite-terrestrial communications. However, as the latency metric is not targeted in the resource allocation optimization among beams, the influence of resource allocation on latency is little, while the throughput of the proposed method yields a marginally superior latency metric performance than that of the benchmark without resource allocation. Moreover, there is a trade-off between the throughput and latency metric. Meanwhile, it is noted that with the increase of β, there is a diminishing growth rate in throughput, while the latency metric becomes significant, and vice versa. Therefore, there is a case of diminishing marginal benefits between the two metrics, which indicates that it is important to choose the proper β to achieve a balance between the throughput and latency metric.

2) Results With Different Data Rates of Input Traffic: The throughput, as well as the latency metric of satellite-terrestrial communications with various data rates of input traffic under $\beta = 0.7$ and $N_s = 16$, is shown in Fig. 7. It can be observed that when the data rate of input traffic is below 22 Gbps, all benchmarks are capable of fulfilling the traffic demands in Fig. 7(a). As the data rate of input traffic increases, the performance disparities among the benchmarks become apparent, while the proposed method is capable of transmitting all input traffic at its data rates below 26 Gbps. Moreover, when the data rate of input traffic exceeds 30 Gbps, the throughput of the proposed method shows an increase of 18.45% compared to that of the original benchmark. Additionally, as mentioned above, it can also be noted that the BH scheduling utilizing the multi-agent DRL, along with resource allocation among beams and load balancing via ISLs, effectively enhances the throughput of satellite-terrestrial communications.

Regarding the latency metric in Fig. 7(b), an initial increase followed by a decrease has been observed as the data rate of input traffic rises. Meanwhile, the latency metric of the benchmark without DRL is similar to that of the proposed method when the input traffic is at low data rates. This phenomenon occurs because all the demands in the queues can always be satisfied at low data rates of input traffic. Thus, only a reduction in $\overline{\tau}_s$ is necessary to decrease the latency metric, while the greedy scheduling for the BH is effective for this. However, when the data rate of input traffic reaches between 26 Gbps and 27 Gbps, the demands of some queues may not be fulfilled due to the non-uniform traffic distribution and time-varying characteristics in LEO satellite constellations. Therefore, the queue length will be unstable, as noted in [12], significantly increasing the latency metric. With a further increase in the data rate of input traffic, the objective of load balancing shifts to equalizing the traffic load among satellites. At this stage, the demand has surpassed

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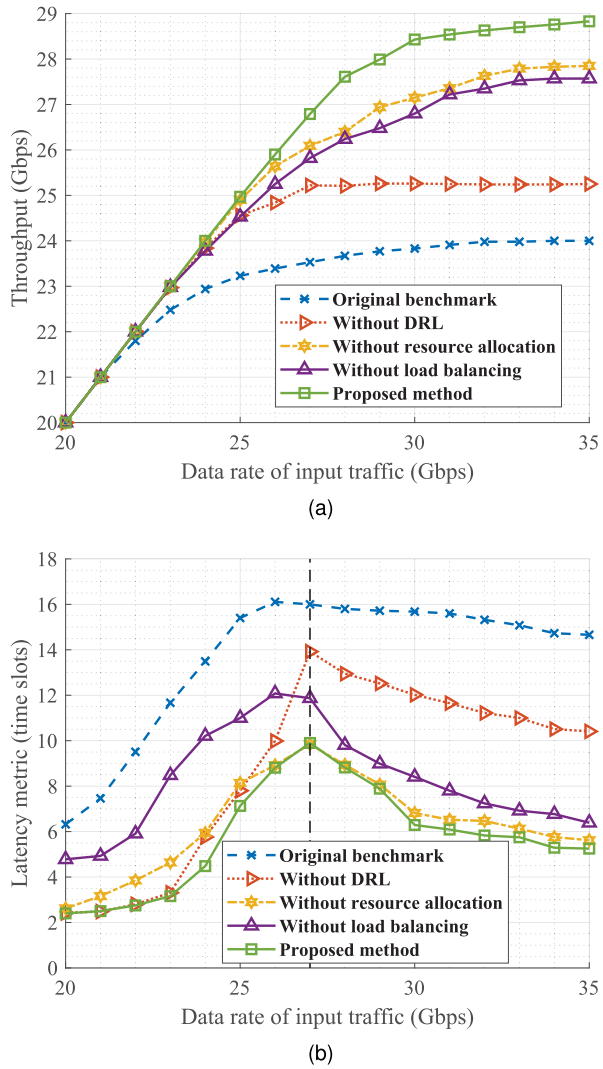

Fig. 7. The performance of different data rates of input traffic versus (a) the throughput and (b) the latency metric, with trade-off coefficient $\beta = 0.7$.

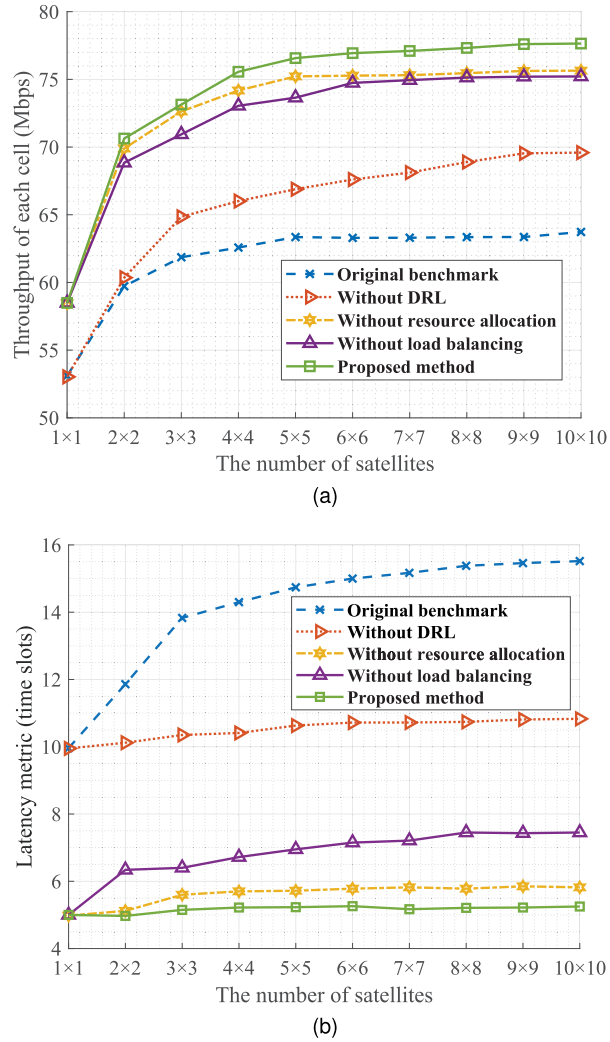

Fig. 8. The performance of different numbers of satellites versus (a) the throughput and (b) the latency metric, with trade-off coefficient β=0.7.

the network capacity, causing congestion in most queues. Therefore, delay fairness among queues is improved, and the latency metric decreases. Moreover, it has also been observed that the load balancing can effectively postpone the peak of the latency metric until the input traffic reaches 27 Gbps, confirming the efficiency of the load balancing via ISLs.

3) Results With Different Numbers of Satellites: Each satellite independently determines the BH pattern by DQN and cooperates with neighboring satellites for resource allocation. Therefore, the computational complexity and additional overhead for each satellite do not increase with the expansion of the constellation size. Moreover, the real-time requirement for updating the load balancing scheme is not high. Consequently, the proposed method can be extended to accommodate larger constellations. The average throughput per cell and the latency metric for the satellite-terrestrial network with different amounts of satellites are depicted in Fig. 8. It can be observed that an increase in satellite amounts leads to a decrease in the average number of cells covered by each satellite, thereby resulting in enhanced throughput of each cell, which suggests that multi-satellite cooperative coverage architecture can effectively improve the throughput of satellite-terrestrial communications, as shown in Fig. 8(a). Moreover, in comparison to other benchmarks, the proposed method consistently outperforms in throughput, maintaining superior performance even as the constellation size expands, which demonstrates the efficiency of the proposed multi-satellite cooperative coverage approach in mega LEO satellite constellations.

For the latency metric illustrated in Fig. 8(b), it is demonstrated that the latency metric of the benchmark without load balancing surges notably as the constellation scale expands, underscoring the significance of load balancing in mega constellations with non-uniform traffic distribution. Meanwhile, the proposed method still outperforms other benchmarks in the latency metric under various constellation scales, which indicates the universality of the proposed method. Furthermore, it can be observed that beyond the 4×4 constellation, the curves in the throughput and latency metric tend to flatten, indicating that the selection of $N_s = 16$ is representative.

VIII. Conclusion

In this paper, a multi-satellite cooperation architecture with BH was proposed for satellite-terrestrial communications.

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Moreover, an optimization problem was formulated to improve the throughput and decrease the latency metric. Then, the overall problem was decomposed into three sub-problems, including the scheduling of the BH with multi-agent DRL, the resource allocation among beams by the MM algorithm, and the load balancing via ISLs under various scenarios. In addition, the proposed method exhibited low computational complexity, thus enabling efficient solution of these sub-problems. Simulation results indicated that the optimization of all three sub-problems can enhance both the throughput and latency metric concurrently. Moreover, the proposed method surpassed other benchmarks with various data rates of input traffic. Specifically, the throughput was improved by 18.45% compared to the original benchmark, with a significant reduction in the latency metric. Additionally, as the constellation scale expanded, the throughput and latency metric of the proposed method remained stable, which indicated its potential application in the future mega LEO constellations.

Appendix A
Proof of Proposition 1

The principal diagonal elements of the Hessian matrix of $\tilde{R}_{s,k}$ with respect to $f_{s,k,l}$ can be calculated as

$$\begin{aligned}\frac{\partial^2 \tilde{R}_{s,k}}{\partial f_{s,k,l}^2} &= \frac{\partial^2 R_{s,k,l}}{\partial f_{s,k,l}^2} \\ &= \frac{B_0}{\ln 2}\frac{(2+\gamma_{s,k,l}(\boldsymbol{f},P_b))}{(1+\gamma_{s,k,l}(\boldsymbol{f},P_b))^2}\frac{\partial \gamma_{s,k,l}(\boldsymbol{f},P_b)}{\partial f_{s,k,l}} > 0. \quad (48)\end{aligned}$$

Therefore, positive eigenvalues exist for this Hessian matrix, which indicates that $\tilde{R}_{s,k}T_0$ is not a concave function, and thus $y_{s,k} - \tilde{R}_{s,k}$ is not a convex function.

Appendix B
Proof of Proposition 2

Note that the example function $w_1(\gamma) = \log_2(1+\gamma)$ exists a lower bound $\underline{w}_1(\gamma)$ as [43]

$$\underline{w}_1(\gamma) = \frac{\gamma^{(q)}}{1+\gamma^{(q)}}\log_2(\gamma) + \log_2(1+\gamma^{(q)}) - \frac{\gamma^{(q)}}{1+\gamma^{(q)}}\log_2(\gamma^{(q)}), \quad (49)$$

and $\underline{w}_1(\gamma^{(q)}) = w_1(\gamma^{(q)})$, $\frac{\mathrm{d}\underline{w}_1}{\mathrm{d}\gamma}|_{\gamma=\gamma^{(q)}} = \frac{\mathrm{d}w_1}{\mathrm{d}\gamma}|_{\gamma=\gamma^{(q)}}$. Therefore, $R_{s,k,l}$ can be approximated as

$$R_{s,k,l} \geq B_0 f_{s,k,l}\Big(\zeta_{s,k,l}^{(q)} - \phi_{s,k,l}^{(q)}\log_2\big(\frac{1}{\gamma_{s,k,l}(\boldsymbol{f},P_b)}\big)\Big). \quad (50)$$

Moreover, for $\log_2(1/\gamma_{s,k,l}(\boldsymbol{f},P_b))$ in the latter term, an example function $w_2(\gamma) = \log_2(1/\gamma)$ is defined, which has an upper bound $\overline{w}_2(\gamma)$ as

$$\overline{w}_2(\gamma) = \gamma^{(q)}\log_2(e)\frac{1}{\gamma} - \log_2(\gamma^{(q)}e), \quad (51)$$

and similarly, $\overline{w}_2(\gamma^{(q)}) = w_2(\gamma^{(q)})$, $\frac{\mathrm{d}\overline{w}_2}{\mathrm{d}\gamma}|_{\gamma=\gamma^{(q)}} = \frac{\mathrm{d}w_2}{\mathrm{d}\gamma}|_{\gamma=\gamma^{(q)}}$. Substituting (51) into (50) gives the result of (26) and verifies the correctness of *Proposition* 2.

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